BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023 • Paper Code: MRD-10TH 120523 (A)

Retyped faithfully from the original board paper — same questions, no changes.

Section – A

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** MRD-10TH 120523 (A)
VERSION: C

Note: Note: There are three sections in this paper i.e. A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Choose the correct option i.e A, B, C or D.

Q1. Natural gas is mainly composed of:

(A) CH_4
(B) C_2H_8
(C) C_2H_6
(D) C_2H_2

Q2. When ethene react with HBr, the compound formed is:

(A) CH_3-CH_2-Br
(B) $Br-CH_2-CH_2-Br$
(C) $CH_3-CH_2-CH_3$
(D) $CH\equiv CH$

Q3. Substitution reaction is the characteristic of:

(A) Alkanes
(B) Alkenes
(C) Alkynes
(D) None of these

Q4. Which one of the following is disaccharide?

(A) Glucose
(B) Fructose
(C) Sucrose
(D) Starch

Q5. The value of Kc is very small it shows that.

(A) Reaction will go in the forward direction.
(B) Reaction will go in the reverse direction.
(C) Reaction is at equilibrium.
(D) Equilibrium will never establish.

Q6. The POH of 0.001M solution of nitric acid is.

(A) 0.001 (B) 10.0

(C) 11 (D) 14.

Q7. The example of weak acid is.

(A) HNO_3 (B) CH_3COOH

(C) Hcl (D) H_2SO_4

Q8. Carbon is bonded on both sides to oxygen atom in.

(A) Alcohol (B) Ketone

(C) Aldehyde (D) Ether

Q9. Deficiency of vitamin-A causes.

(A) Scurvy (B) Rickets

(C) Night Blindness (D) Aging

Q10. All are air pollutants except:

(A) No_2 (B) N_2

(C) SO_2 (D) SO_3

Q11. The PH of acid rain is less than.

(A) 13.6 (B) 10.6

(C) 7.6 (D) 5.6

Q12. Matte is a mixture of:

(A) CuS and FeS (B) Cu_2S and FeS

(C) Cu_2O and FeS (D) Cu_2O and FeO

Section – B

Time Allowed: 2.40 Hours | **Maximum Marks:** 32

Q2. Write short answers to any eight (8) of the following parts. All parts carry equal marks:-

(i) Write the forward and reverse reaction of the following reversible reaction:- $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$

(ii) How would you identify that dynamic equilibrium is established?

(iii) Justify H^+ ion as a Lewis acid?

(iv) Calculate the pH of 0.2M solution of NaOH.

(v) What are cycloalkanes? Give two examples.

(vi) Compare the reactivity of alkane and alkene.

(vii) Write down four uses of lipids.

(viii) What are the adverse effects of global warming?

(ix) Give four disadvantages of the detergents.

(x) Enlist at least four (4) uses of Urea.

(xi) What is slaked lime? How slaked lime is produced?

Section – C

Maximum Marks: 21

> Note: Answer any three (3) questions. All questions carry equal marks:-

Q3.

(a) How can you predict the direction of reaction from KC value?

(b) Discuss the concept of Lewis acids and bases with examples.

Q4.

(a) List the different characteristics of organic compounds.

(b) Write down the equations for the preparation of alkanes, alkenes and alkynes.

Q5.

(a) What functions do carbohydrates perform in living organisms?

(b) Explain ozone layer depletion.

Q6.

(a) Make the distinction between soft and hard water.

(b) Sketch the flow sheet diagram of the solvay process.